



- 2 (a) Starting from the definitions of  $\tanh$  and  $\operatorname{sech}$  in terms of exponentials, prove that

$$\tanh^2 t + \operatorname{sech}^2 t = 1. \quad [3]$$

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- (b) The curve  $C$  has parametric equations

$$x = \ln(\cosh t), \quad y = \tan^{-1}(\sinh t), \quad \text{for } 0 \leq t \leq 1.$$

Find the length of  $C$ . [5]

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